

Step 1: Input file preparation

- Generate the 'FORCE_CONSTANTS' file (supported formats: phonopy or Quantum ESPRESSO)
- Prepare the configuration file 'input.txt'. The **Input.txt template** can also be found in “./docs/Dysurf_online_version_manual/” directory.

Input.txt template

! Example of SQE input file for Fe₃GaTe₂

&basic

ntypes = 3

natoms = 12

nsite = 3 3 2

/

&inputsqe

elements = "Fe" "Ga" "Te"

nat = 6 2 4

clatvec(:,1) = 4.0472469330 0.0000000000 0.0000000000

clatvec(:,2) = 0.0000000000 7.0100378990 0.0000000000

clatvec(:,3) = 0.0000000000 0.0000000000 16.3504714966

temp = 300.d0

lneutron = .TRUE.

lxray = .FALSE.

qmesh = 20 20 20

write_rmsd = .TRUE.

!read_rmsd = .TRUE.

!espresso = .TRUE. path(:,1)

= 0 0 1

!path(:,2) = 0 1 0

!path(:,3) = 0 0 1

ne = 400

deltaE = 0.08

nqh = 100

nqk = 2

nql = 2

deltaH = 0.005

deltaK = 0.01

deltaL = 0.01

q0 = AAA

!aff_wk = .TRUE.

!aff_a(:,1) = 31.066358 15.341823 49.278296 4.577665 16.828321

!aff_a(:,2) = 6.372157 5.154568 1.473732 1.635073 1.209372

!aff_b(:,1) = 1.708732 9.618455 0.000760 66.146202 0.168002

!aff_b(:,2) = 1.514347 22.092528 0.061373 55.445176 0.646925

!aff_c(:) = -44.119025 0.154722

nonanalytic = .FALSE.

lresfunc = .TRUE.

```
functype = "CNCS12"
```

```
xm = 2
```

```
!ltds=.TRUE.
```

```
lphase=.FALSE.
```

```
/
```

LATTICE_PARAMETERS

```
1.0000000000000000
```

```
4.0472474720811968 0.0000000000000000 0.0000000000000000
```

```
-2.0236237360405984 3.5050191262119252 -0.0000000000000000
```

```
0.0000000000000000 -0.0000000000000000 16.3504713097626571
```

ATOMIC_POSITIONS

Direct

```
0.0000000000000000 0.0000000000000000 0.1762438733901837
```

```
-0.0000000000000000 0.0000000000000000 0.3237561266098163
```

```
-0.0000000000000000 0.0000000000000000 0.6762438733901838
```

```
-0.0000000000000000 0.0000000000000000 0.8237561266098162
```

```
0.6666666870000029 0.3333333429999996 0.7500000000000000
```

```
0.3333333129999971 0.6666666269999979 0.2500000000000000
```

```
0.3333333429999996 0.6666666870000029 0.7500000000000000
```

```
0.6666666269999979 0.3333333129999971 0.2500000000000000
```

```
0.6666666870000029 0.3333333429999996 0.5944189348263391
```

```
0.3333333129999971 0.6666666269999979 0.4055810651736609
```

```
0.3333333129999971 0.6666666269999979 0.0944189348263390
```

```
0.6666666870000029 0.3333333429999996 0.9055810651736609
```

Step 2: Upload files

After preparing both input.txt and the force constants file, upload them on the website (<http://36.138.185.163:5000/>).

The screenshot shows a web interface for uploading files. It has a purple header with an 'Upload' button. Below it, there are two dashed boxes for file selection. The first box is labeled '1: Input Parameter' and 'step1' in red. It contains a 'Choose file' button and a text field showing 'No file'. Below the text field, it says 'Parameter files (.txt, .in, .inp, .dat formats supported)'. The second box is labeled '2: FORCE_CONSTANTS (Phonopy/QE format)' and 'step2' in red. It also contains a 'Choose file' button and a text field showing 'No file'. Below the text field, it says 'FORCE_CONSTANTS file (supports large files up to 100MB)'. At the bottom, there is a green button labeled 'Upload' and 'step3' in red.

Step 3: Execution

